Exercise1: INTRODUCTION

1. **Introduction to Python libraries for Data Mining: NumPy, Pandas, Matplotlib etc**.

Write a Python program to do the following operations: Library: NumPy

a) Create multi-dimensional arrays and find its shape and dimension
b) Create a matrix full of zeros and ones
c) Reshape and flatten data in the array
d) Append data vertically and horizontally
e) Apply indexing and slicing on array
f) Use statistical functions on array - Min, Max, Mean, Median and Standard Deviation

PROCEDURE:

1. Create: Open a new file in Python shell, write a program and save the program with .py extension.
2. Execute: Go to Run -> Run module (F5)

a) Create multi-dimensional arrays and find its shape and dimension

```
import numpy as np
#creation of multi-dimensional array
a=np.array([[1,2,3],[2,3,4],[3,4,5]])
#shape
b=a.shape
print("shape:")
print(a.shape)
#dimension
c=a.ndim
print("dimensions:")
print(a.ndim)
```

b) Create a matrix full of zeros and ones

```
import numpy as np
 #matrix full of zeros
z=np. zeros ((2,2))
print("zeros:")
```

```
print(z)
```

#matrix full of ones

```
o=np. ones ((2,2))
print("ones:")
print(o)
```

c) Reshape and flatten data in the

```
import numpy as np
a=np.array([[1,2,3,4],[2,3,4,5],[3,4,5,6],[4,5,6,7]])
b=a.reshape(4,2,2)
print("reshape:")
print(b)
#matrix flatten
c=a.flatten()
print("flatten:")
print(c)
```

d) Append data vertically and horizontally

```
import numpy as np
#Appending data vertically
x=np.array([[10,20],[80,90]])
y=np.array([[30,40],[60,70]])
v=np.vstack((x,y))
print("vertically:")
print(v)
#Appending data horizontally
h=np.hstack((x,y))
print("horizontally:")
print(h)
```

e) Apply indexing and slicing on array #indexing

```
import numpy as np
a=np.array([[1,2,3,4],[2,3,4,5],[3,4,5,6],[4,5,6,7]])
```

```
temp = a[[0, 1, 2, 3],[1, 1, 1, 1]]
print('indexing:')
print(temp)
#slicing
i=a[:4,::2]
print('slicing:')
print(i)
```

f) Use statistical functions on array - Min, Max, Mean, Median and Standard Deviation

```
import numpy as np
#min for finding minimum of an array
a=np.array([[1,3,-1,4],[3,-2,1,4]])
b=a.min()
print("minimum:",b)
#max for finding maximum of an array
c=a.max()
print("maximum:",c)
#mean
a=np.array([1,2,3,4,5])
d=a.mean()
print("mean:",d)
#median
e=np.median(a)
print("median:",e)
#standard deviation
f=a.std()
print("standard deviation:",f)
```